Mini Project 3 - Text Classification

Classifying mental health comments into categories

Importing libararies to conduct text classification

In [234…

```
## Import Libraries
import numpy as np
import pandas as pd

import string
import spacy
import regex as re
import contractions

%matplotlib inline
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import seaborn as sns
from wordcloud import WordCloud, STOPWORDS

from collections import Counter

from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.decomposition import LatentDirichletAllocation
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB, ComplementNB
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import GaussianNB
from sklearn.pipeline import Pipeline
from sklearn.metrics import classification_report

import warnings
warnings.filterwarnings('ignore')
```

Exploratory Data Analysis

In [394…

```
#Reading csv file into dataframe
df = pd.read_csv('mental_health_combined_test.csv')
```

```python
df.sample(10)
```

Out [394…

	text	status
352	sounds weird, right? but lately i've been so d...	Depression
193	(in informal settings when there is sun obviou...	Anxiety
242	i don’t know if this has been posted on this t...	Anxiety
778	I've just about had it.Life is short. Too shor...	Suicidal
897	People who would Care if I diedThe only people...	Suicidal
577	All lives can't matter until black lives matte...	Normal
112	my wife passed out from a panic attack yesterd...	Anxiety
203	which is, now, the longest i've gone without a...	Anxiety
605	Hello there My name Ragnikant Babu From Indian...	Normal
701	I've got suspended for a joke and I'm pissed W...	Normal

In [395…

```python
#value count for status
df['status'].value_counts()
```

Out [395…

```
status
Anxiety       248
Depression    248
Normal        248
Suicidal      248
Name: count, dtype: int64
```

In [396…

```python
#Applying cleaning function to text data
def clean_text(text):
    # reduce multiple spaces and newlines to only one
    text = re.sub(r'(\s\s+|\n\n+)', r'\1', text)
    # remove double quotes
    text = re.sub(r'"', '', text)

    return text

df['text'] = df['text'].apply(clean_text)
```

In [397…

```python
df.head()
```

Out [397…

	text	status
0	i don't understand whats wrong with me. i don'...	Anxiety
1	usually when i have anxiety just chatting with...	Anxiety
2	well, i've had anxiety and panic syndrome for ...	Anxiety
3	for the most minimal of things, like standing ...	Anxiety
4	i stay away from family and live with my roomm...	Anxiety

In [398…
```
df['text'] = df['text'].apply(lambda x: contractions.fix(str(x)))
```

In [399…
```
df.sample(10)
```

Out [399…

	text	status
314	you are depressed! you are okay, it will take ...	Depression
220	i ordered a pizza online but after i submitted...	Anxiety
248	i am talking about the kind of people who woul...	Depression
780	I have considered suicide, but realized it wou...	Suicidal
731	Who. Wants. My. Goddamn. Hugz. Award Say somet...	Normal
759	:(i literally cannot remember the last time i ...	Suicidal
227	for example: let us say i have this appointmen...	Anxiety
974	I cannot live like this anymore.I am dying ins...	Suicidal
104	making friends is something i am legitimately ...	Anxiety
452	whether it is people, places or phases in life...	Depression

In [400…
```
nlp = spacy.load('en_core_web_sm')
```

In [401…
```
#function to remove stop words, numbers, and punctuation, while retain
# and lemmatizing words
def convert_text(text):
    sent = nlp(text)
    ents = {x.text: x for x in sent.ents}
    tokens = []
    for w in sent:
        if w.is_stop or w.is_punct or w.is_digit:
            continue
        if w.text in ents:
            tokens.append(w.text)
        else:
            tokens.append(w.lemma_.lower())
    text = ' '.join(tokens)
```

```
    return text
```

In [402…

```
%%time
df['short'] = df['text'].apply(convert_text)
```

```
CPU times: user 21 s, sys: 1.78 s, total: 22.8 s
Wall time: 22.9 s
```

In [403…

```
df["short"] = df["short"].str.replace(
    r"[^a-zA-Z0-9 ]", " ", regex=True
)
```

In [405…

```
df.sample(10)
```

Out[405…

	text	status	short
346	i pretty much do not know where to go at this ...	Depression	pretty know point tell thing bad thing actuall...
430	i usually do not wash or fold unless i absolut...	Depression	usually wash fold absolutely typically look li...
349	after having to take a medical leave from a go...	Depression	have medical leave good college depression soc...
303	this is becoming a problem. this is not the fi...	Depression	problem time happen surprised fire unmotivated...
974	I cannot live like this anymore.I am dying ins...	Suicidal	live like anymore die inside go right anymore ...
307	i did not wake up yesterday until 6pm. i laid ...	Depression	wake yesterday pm lay bed wake buy case beer p...
616	There were a lot of moms protesting quarintine...	Normal	lot mom protest quarintine area start quarinti...
867	Is this legitIn comments is the link. If I bou...	Suicidal	legitin comment link buy take actually kill
334	my depression has the opposite effect on me, i...	Depression	depression opposite effect make want eat lot t...
623	my judgement has finally come... according to ...	Normal	judgement finally come accord r teenamiugly ...

In [406…

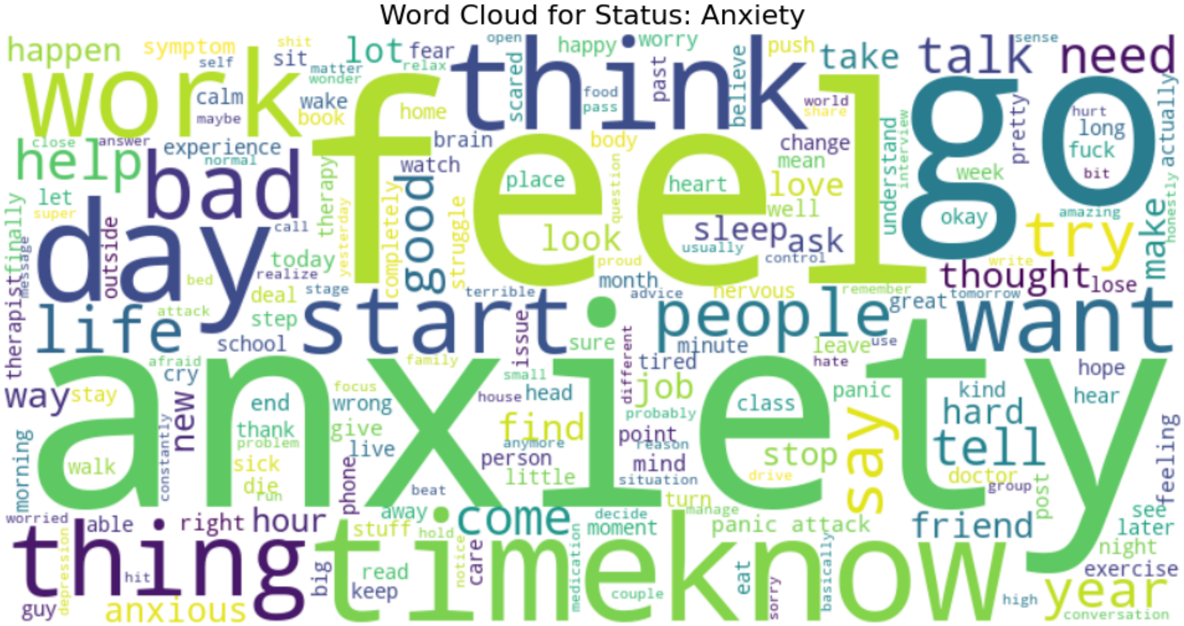
```
#filtering data by target status
#setting targets
target_anxiety = 'Anxiety'
target_depression ='Depression'
target_normal ='Normal'
target_suicidal ='Suicidal'

#Creating df by target status
```

```
anxiety_df = df[df['status'] == target_anxiety]
depression_df = df[df['status'] == target_depression]
normal_df = df[df['status'] == target_normal]
suicidal_df = df[df['status'] == target_suicidal]

#Joining all rows into one cell
anxiety_data = " ".join(anxiety_df["short"].dropna().astype(str))
depression_data = " ".join(depression_df["short"].dropna().astype(str)
normal_data = " ".join(normal_df["short"].dropna().astype(str))
suicidal_data = " ".join(suicidal_df["short"].dropna().astype(str))
```

In [407…

```
# Configure and generate the word cloud for anxiety
# Adding default STOPWORDS filters out common filler words like 'and',
wordcloud_anxiety = WordCloud(
    width=800,
    height=400,
    background_color="white",
    stopwords=set(STOPWORDS),
    colormap="viridis",
    min_font_size=10,
).generate(anxiety_data)

#Display the final visualization using Matplotlib
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_anxiety, interpolation="bilinear")
plt.axis("off")  # Hides x and y axis coordinates for a clean visual
plt.title(f"Word Cloud for Status: {target_anxiety.capitalize()}", fon
plt.tight_layout(pad=0)
plt.show()
```

Word Cloud for Status: Anxiety



In [408…

```
#Finding frequency of words for anxiety data
anxiety_data_split = anxiety_data.split()
anxiety_data_count = Counter(anxiety_data_split)
anxiety_data_top = anxiety_data_count.most_common(10)
```

```
anxiety_data_top

#creating dataframe for top 10 words
anxiety_data_top = pd.DataFrame(anxiety_data_top, columns=['Word', 'Co
anxiety_data_top.index = ['1', '2', '3', '4', '5', '6', '7', '8', '9',
anxiety_data_top
```

Out[408…

	Word	Count
1	anxiety	340
2	feel	318
3	like	286
4	go	210
5	time	205
6	know	194
7	think	163
8	day	157
9	work	149
10	thing	147

In [409…

```
# Configure and generate the word cloud for depression
# Adding default STOPWORDS filters out common filler words like 'and',
wordcloud_depression = WordCloud(
    width=800,
    height=400,
    background_color="white",
    stopwords=set(STOPWORDS),
    colormap="viridis",
    min_font_size=10,
).generate(depression_data)

# Display the final visualization using Matplotlib
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_depression, interpolation="bilinear")
plt.axis("off")  # Hides x and y axis coordinates for a clean visual
plt.title(f"Word Cloud for Status: {target_depression.capitalize()}",
plt.tight_layout(pad=0)
plt.show()
```

Word Cloud for Status: Depression

In [365…

```python
#Finding frequency of words for depression data
depression_data_split = depression_data.split()
depression_data_count = Counter(depression_data_split)
depression_data_top = depression_data_count.most_common(10)

#creating dataframe for top 10 words
depression_data_top = pd.DataFrame(depression_data_top, columns=['Word
depression_data_top.index = ['1', '2', '3', '4', '5', '6', '7', '8', '
depression_data_top
```

Out[365…

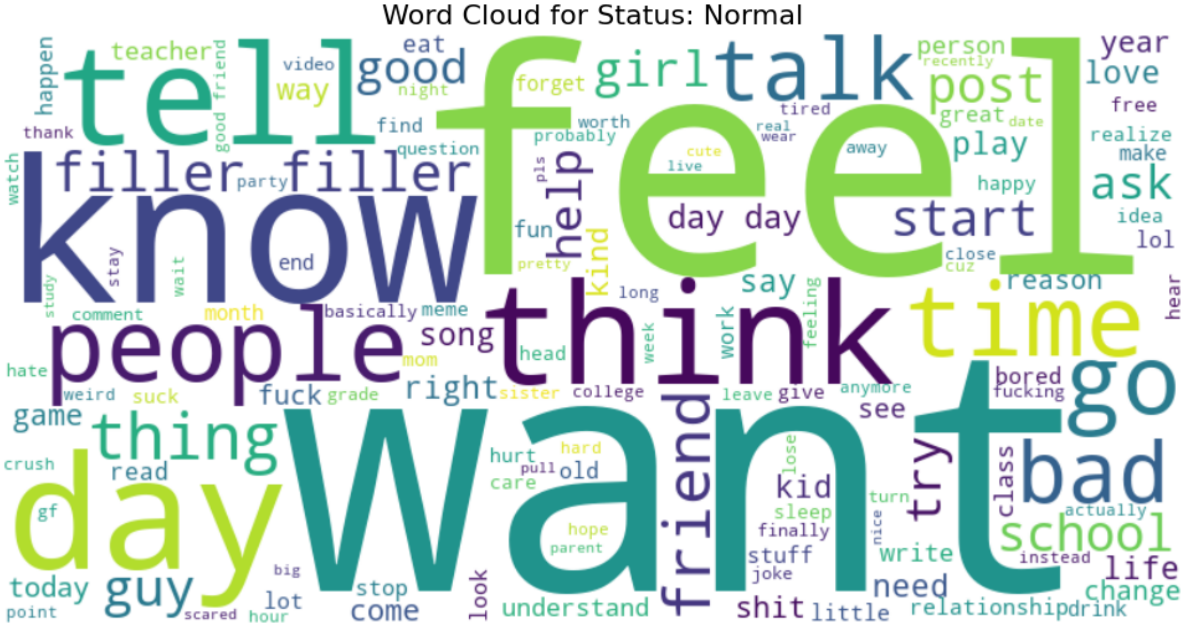
	Word	Count
1	feel	311
2	like	254
3	want	192
4	know	164
5	people	162
6	life	162
7	time	158
8	go	144
9	day	142
10	think	136

In [366…

```python
# Configure and generate the word cloud for normal
# Adding default STOPWORDS filters out common filler words like 'and',
wordcloud_normal = WordCloud(
```

```
    width=800,
    height=400,
    background_color="white",
    stopwords=set(STOPWORDS),
    colormap="viridis",
    min_font_size=10,
).generate(normal_data)

# Display the final visualization using Matplotlib
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_normal, interpolation="bilinear")
plt.axis("off")  # Hides x and y axis coordinates for a clean visual
plt.title(f"Word Cloud for Status: {target_normal.capitalize()}", font
plt.tight_layout(pad=0)
plt.show()
```

Word Cloud for Status: Normal



In [367…

```
#Finding frequency of words for normal data
normal_data_split = normal_data.split()
normal_data_count = Counter(normal_data_split)
normal_data_top = normal_data_count.most_common(10)

#creating dataframe for top 10 words
normal_data_top = pd.DataFrame(normal_data_top, columns=['Word', 'Coun
normal_data_top.index = ['1', '2', '3', '4', '5', '6', '7', '8', '9',
normal_data_top
```

Out [367…

	Word	Count
1	like	137
2	day	121
3	want	89
4	feel	69
5	know	68
6	friend	51
7	think	50
8	tell	48
9	people	43
10	get	43

In [368…

```
# Configure and generate the word cloud for Suicidal
# Adding default STOPWORDS filters out common filler words like 'and',
wordcloud_suicidal = WordCloud(
    width=800,
    height=400,
    background_color="white",
    stopwords=set(STOPWORDS),
    colormap="viridis",
    min_font_size=10,
).generate(suicidal_data)

# Display the final visualization using Matplotlib
plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_suicidal, interpolation="bilinear")
plt.axis("off")  # Hides x and y axis coordinates for a clean visual
plt.title(f"Word Cloud for Status: {target_suicidal.capitalize()}", fo
plt.tight_layout(pad=0)
plt.show()
```

Word Cloud for Status: Suicidal


```
In [369…
#Finding frequency of words for anxiety data
suicidal_data_split = suicidal_data.split()
suicidal_data_count = Counter(suicidal_data_split)
suicidal_data_top = suicidal_data_count.most_common(10)

#creating dataframe for top 10 words
df_suicidal_data_top = pd.DataFrame(suicidal_data_top, columns=['Word'
df_suicidal_data_top.index = ['1', '2', '3', '4', '5', '6', '7', '8',
df_suicidal_data_top
```

Out[369…

	Word	Count
1	want	360
2	like	323
3	know	275
4	feel	272
5	life	262
6	think	212
7	go	193
8	people	183
9	time	180
10	get	153

Feature Engineering keywords: Anxiety, Suicide, Depression

```
In [411…
keywords = ['anxiety', 'anxious', 'suicide', 'suicidal', 'depress', 'd
```

```
for kw in keywords:
    df[f'flag_{kw}'] = df['short'].str.contains(kw, case=False, na=Fal
```

Applying Text Classifcation models

In [371…

```
#Previewing new dataframe
df.sample(5)
```

Out[371…

	text	status	short	flag_anxiety	flag_anxious	flag_su
231	my therapist said that i did not need him anym...	Anxiety	therapist say need anymore say improve bi mont...	1	0	
659	Just did a pact with my best friend If neither...	Normal	pact good friend married relationship age adop...	0	0	
224	the thought of my spouse dying is the main thi...	Anxiety	thought spouse dying main thing give high anxi...	1	0	
936	please tell me how to kill myself.please tell ...	Suicidal	tell kill myselfplease tell reliably kill pain...	0	0	
482	i make a lot of mistakes and am very awkward s...	Depression	lot mistake awkward end think word kill dozens...	0	0	

Setting features and labels using cleaned_short text

In [372…

```
# Setting Features and Labels using cleaned_short text
X = df['short']
y = df['status']

# Apply a train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size =
```

In [373…

```
X_train
```

Out[373…
```
938    give lifeive life month life actually beautifu...
572    guy sixth grade class convince jack rub jaw mu...
926    slowly happy suicide depression supress gym st...
404    feel like lately terrify tell head true merely...
526    life disappointment  play ball foot game lift ...
                              ...
106    find matter write reply post social medium hon...
270    depression little year go med therapist life c...
860    go end sooni know try kill like month ago fail...
435    take number ssris depressive episode wellbutri...
102    think have great time have fun regret say hate...
Name: short, Length: 694, dtype: object
```

In [374…
```
y_train
```

Out[374…
```
938      Suicidal
572        Normal
926      Suicidal
404    Depression
526        Normal
          ...
106       Anxiety
270    Depression
860      Suicidal
435    Depression
102       Anxiety
Name: status, Length: 694, dtype: object
```

Creating Pipelines to implement classification models

In [375…
```
pipeMNB = Pipeline([('tfidf', TfidfVectorizer(ngram_range=(1, 3), min_
pipeCNB = Pipeline([('tfidf', TfidfVectorizer(ngram_range=(1, 3), min_
pipeRF = Pipeline([('tfidf', TfidfVectorizer(ngram_range=(1, 3), min_d
pipeGBC = Pipeline([('tfidf', TfidfVectorizer(ngram_range=(1, 3), min_
pipeSVC = Pipeline([('tfidf', TfidfVectorizer(ngram_range=(1, 3), min_
```

In [376…
```
pipeMNB.fit(X_train, y_train)
predictMNB = pipeMNB.predict(X_test)
accuracy_score_MNB = accuracy_score(y_test, predictMNB)
```

In [377…
```
pipeRF.fit(X_train, y_train)
predictRF = pipeRF.predict(X_test)
accuracy_score_RF = accuracy_score(y_test, predictRF)
```

In [378…
```
pipeGBC.fit(X_train, y_train)
pipeGBC = pipeGBC.predict(X_test)
accuracy_score_GBC = accuracy_score(y_test, pipeGBC)
```

In [379…
```
pipeSVC.fit(X_train, y_train)
pipeSVC = pipeSVC.predict(X_test)
accuracy_score_SVC = accuracy_score(y_test, pipeSVC)
```

In [380…
```
pipeCNB.fit(X_train, y_train)
predictCNB = pipeCNB.predict(X_test)
accuracy_score_CNB = accuracy_score(y_test, predictCNB)
```

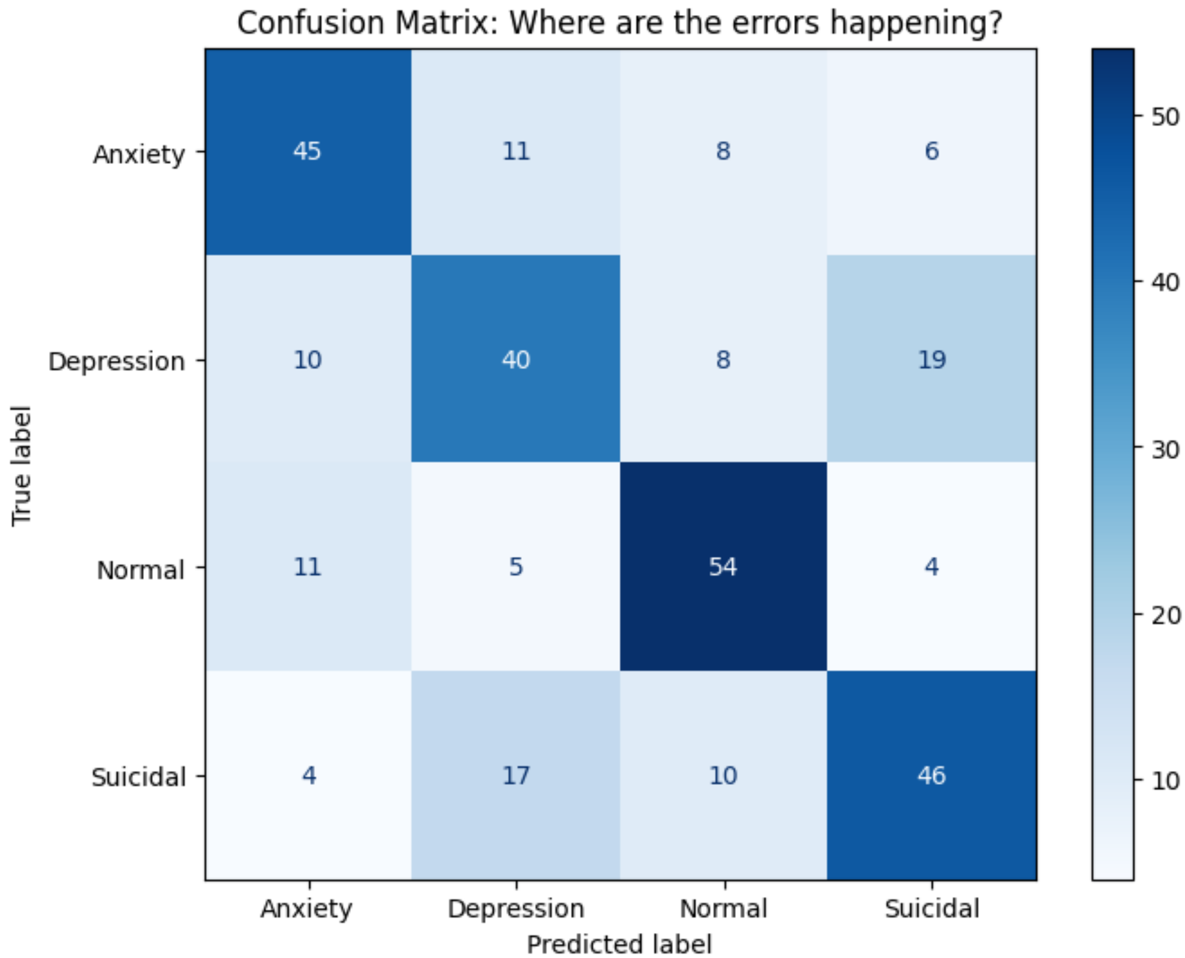
In [381…
```
print(f"MultinomialNB: {accuracy_score_MNB:.3f}")
print(f"RandomForestClassifier: {accuracy_score_RF:.3f}")
print(f"GradientBoostingClassifier: {accuracy_score_GBC:.3f}")
print(f"LinearSVC: {accuracy_score_SVC:.3f}")
print(f"ComplementNB: {accuracy_score_CNB:.3f}")
```

```
MultinomialNB: 0.550
RandomForestClassifier: 0.570
GradientBoostingClassifier: 0.631
LinearSVC: 0.641
ComplementNB: 0.577
```

In [382…
```
#Creating classification report for SVC model
print(classification_report(y_test, pipeSVC))
```

```
              precision    recall  f1-score   support

     Anxiety       0.66      0.66      0.66        70
  Depression       0.58      0.53      0.55        77
      Normal       0.70      0.76      0.73        74
    Suicidal       0.62      0.62      0.62        77

    accuracy                           0.64       298
   macro avg       0.64      0.64      0.64       298
weighted avg       0.64      0.64      0.64       298
```

In [176…
```
from sklearn.metrics import ConfusionMatrixDisplay

# Generate and plot the matrix
fig, ax = plt.subplots(figsize=(8, 6))
ConfusionMatrixDisplay.from_predictions(
    y_test,
    pipeSVC,
    display_labels=['Anxiety', 'Depression', 'Normal', 'Suicidal'],
    cmap=plt.cm.Blues,
    ax=ax
)

plt.title("Confusion Matrix: Where are the errors happening?")
plt.show()
```



Setting features and labels using cleaned_short text and feature engineered variables

In [412…

```
#Previewing new dataframe
df.sample(3)
```

Out [412...

	text	status	short	flag_anxiety	flag_anxious	flag_suicide
659	Just did a pact with my best friend If neither...	Normal	pact good friend married relationship age adop...	0	0	0
277	the most popular girl in our entire school her...	Depression	popular girl entire school go brag depressed c...	0	0	0
528	Does anyone need to talk I feel like playing t...	Normal	need talk feel like play therapist right want ...	0	0	0

In [413...

```
# Setting Features and Labels using cleaned_short text
X2 = df[['short','flag_anxiety','flag_anxious','flag_suicide','flag_su
y2 = df['status']

# Apply a train-test split
X2_train, X2_test, y2_train, y2_test = train_test_split(X2, y2, test_s
```

In [414...

```
X2_train
```

Out [414...

	short	flag_anxiety	flag_anxious	flag_suicide	flag_suicidal	flag
938	give lifei ve life month life actually beautif...	0	0	0	0	
572	guy sixth grade class convince jack rub jaw mu...	0	0	0	0	
926	slowly happy suicide depression supress gym st...	0	0	1	0	

404	feel like lately terrify tell head true merely...	0	0	0	0
526	life disappointment play ball foot game lift...	0	0	0	0
...	...	...	...	...	...
106	find matter write reply post social medium hon...	1	0	0	0
270	depression little year go med therapist life c...	0	0	0	0
860	go end sooni know try kill like month ago fail...	0	0	1	0
435	take number ssris depressive episode wellbutri...	0	0	0	0
102	think have great time have fun regret say hate...	0	0	0	0

694 rows × 7 columns

In [415…

```
y2_train
```

Out[415…

```
938      Suicidal
572        Normal
926      Suicidal
404    Depression
526        Normal
          ...
106       Anxiety
270    Depression
860      Suicidal
435    Depression
102       Anxiety
Name: status, Length: 694, dtype: object
```

In [416…
```
numeric_features = ['flag_anxiety','flag_anxious','flag_suicide','flag
                    'flag_depress','flag_depression']
```

In [417…
```
preprocessor = ColumnTransformer(
    transformers=[
        ('text_tfidf', TfidfVectorizer(ngram_range=(1, 3)), 'short'),
        ('numeric_scaler', StandardScaler(), numeric_features)
    ])
```

In [418…
```
pipeSVC2 = Pipeline([('preprocessor', preprocessor), ('clf', LinearSVC
pipeRF2 = Pipeline([('preprocessor', preprocessor), ('clf', RandomFore
pipeGBC2 = Pipeline([('preprocessor', preprocessor), ('clf', GradientB
pipeGNB2 = Pipeline([('preprocessor', preprocessor), ('clf', GaussianN
```

In [419…
```
pipeSVC2.fit(X2_train, y2_train)
pipeSVC_2 = pipeSVC2.predict(X2_test)
accuracy_score_SVC_2 = accuracy_score(y2_test, pipeSVC_2)
print(f"LinearSVC: {accuracy_score_SVC_2:.3f}")
```

```
LinearSVC: 0.644
```

In [420…
```
pipeRF2.fit(X2_train, y2_train)
pipeRF_2 = pipeRF2.predict(X2_test)
accuracy_score_RF_2 = accuracy_score(y2_test, pipeRF_2)
print(f"RandomForestClassifier: {accuracy_score_RF_2:.3f}")
```

```
RandomForestClassifier: 0.584
```

In [421…
```
pipeGBC2.fit(X2_train, y2_train)
pipeGBC_2 = pipeGBC2.predict(X2_test)
accuracy_score_GBC_2 = accuracy_score(y2_test, pipeGBC_2)
print(f"GradientBoostingClassifier: {accuracy_score_GBC_2:.3f}")
```

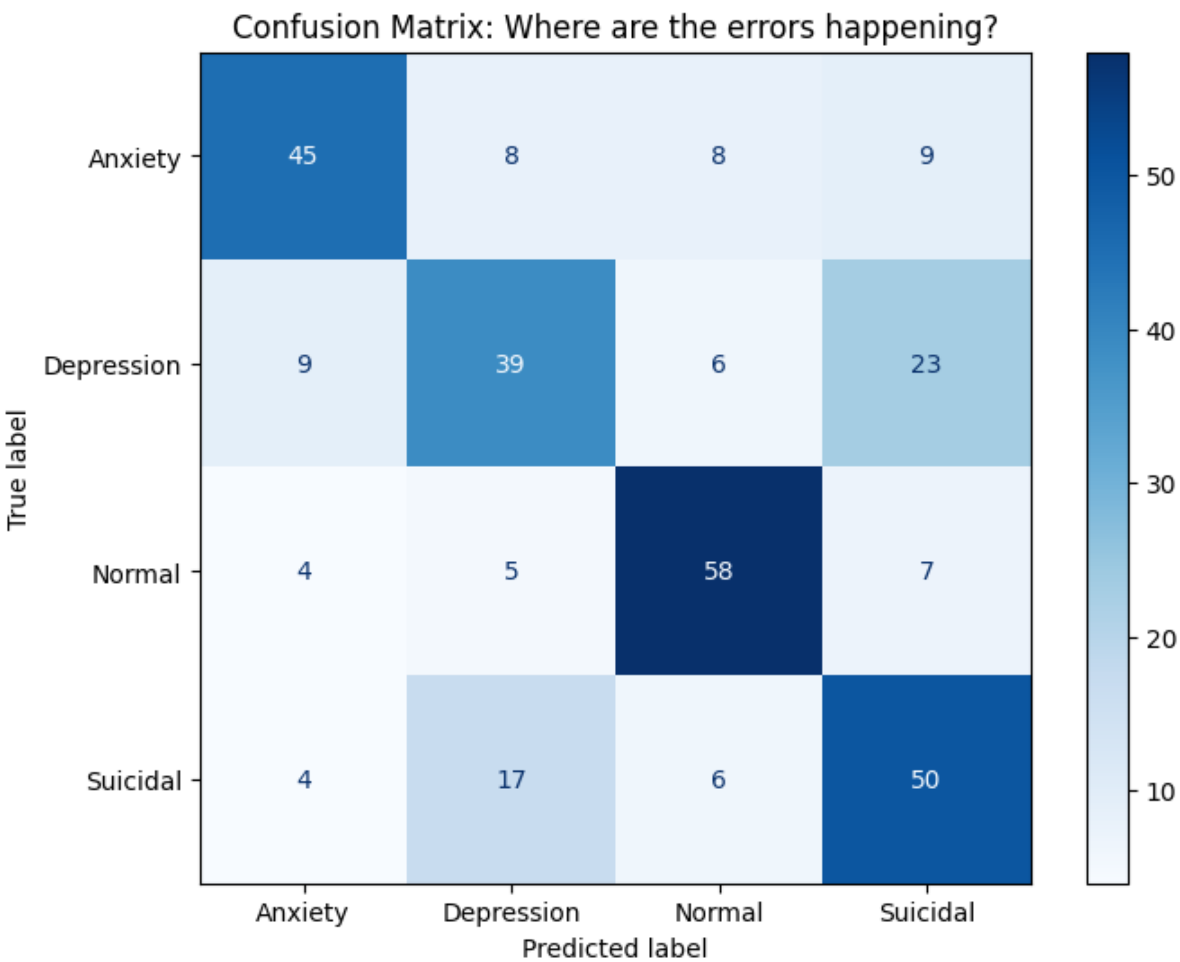
```
GradientBoostingClassifier: 0.624
```

In [422…
```
#Creating classification report for SVC model
print(classification_report(y2_test, pipeSVC_2))
```

```
              precision    recall  f1-score   support

     Anxiety       0.73      0.64      0.68        70
  Depression       0.57      0.49      0.53        77
      Normal       0.73      0.80      0.76        74
    Suicidal       0.57      0.65      0.61        77

    accuracy                           0.64       298
   macro avg       0.65      0.65      0.64       298
weighted avg       0.64      0.64      0.64       298
```

In [194…
```
# Generate and plot the matrix
fig, ax = plt.subplots(figsize=(8, 6))
ConfusionMatrixDisplay.from_predictions(
```

```
    y2_test,
    pipeSVC_2,
    display_labels=['Anxiety', 'Depression', 'Normal', 'Suicidal'],
    cmap=plt.cm.Blues,
    ax=ax
)

plt.title("Confusion Matrix: Where are the errors happening?")
plt.show()
```



Conclusion

- The best performing model for text classification was Linear Support Vector Machines with an accuracy score of 64.4%, using feature engineering.
- Using feature engineering only added minor improvement (0.3%) With only an 64.4% accuracy score and the consequences of misdiagnosis, I would not recommend any of the text classification models currently designed
- Next Steps: Create other features such as word count, word density, character count, etc. Use other text cleaning methods:

- Keeping stop words may help capture nuances of mental health status